# Mechanism-class clinical track record, not target affinity, predicts cognition-drug repurposing success

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## Abstract

Computational drug repurposing for cognitive impairment is dominated by target-centric paradigms: ranking candidates by predicted target-binding affinity (drug–target-interaction foundation models) and by target genetic evidence (Open Targets). On a leakage-audited ledger of 31 cognition drugs with adjudicated pivotal-trial outcomes, we show that **a drug's mechanism-class clinical track record dominates every target-level predictor** of clinical success. The comparison is the headline. On the identical drugs, no target-level predictor beats chance: target affinity scores AUROC 0.47 and target genetic relevance 0.53; the two comparators that appear to work — a knowledge-graph network-propagation score and a chemical-structure-similarity score, both 0.80 — do so only through hindsight confounding. A mechanism-class prognostic prior, predicting a held-out drug's outcome from its class siblings' meta-analytic record, instead separates SUCCESS from FAILURE completely (label-permutation p = 0.0002; 0 of 5000 permutations matched). This perfect separation is **not a generalisable predictive feature** but a direct readout of mechanism-class outcome homogeneity: all 11 classes in the ledger are outcome-pure (uniformly SUCCESS or FAILURE), so the prior is effectively a historical class look-up. A class-level (cluster) bootstrap and an explicit leave-one-class-out ceiling (AUROC 0.00) bound the claim; the method quantifies historical class destiny, it does not forecast a novel mechanism. Crucially the signal is **pseudo-prospective and granularity-specific**: in a prequential "as-of" evaluation predicting each drug from only strictly-earlier drugs, the prior would have flagged the 2014–2022 failure wave before it read out (AUROC 1.00 over drugs with a class precedent, 0.96 overall, with one honest miss), while a coarser taxonomy drops the AUROC to 0.62 and random groupings to 0.46 (0 of 2000 reaching 1.00). The advantage survives restriction to a common subset where every predictor is defined, and a per-disease reframe recovers each disease's real winning mechanism — Alzheimer's (cholinesterase inhibitors; within-disease AUROC 0.97), schizophrenia-associated cognitive impairment (muscarinic M1/M4) and Fragile X (PDE4). Separately, the released MAMMAL DTI head cannot rank ligands within a target at allosteric/transporter sites (leave-one-target-out Spearman ρ −0.12 over 297 ChEMBL pairs), a gap inexpensive cheminformatics features close. Cognition repurposing should be class-aware, not affinity-driven.

**Keywords:** drug repurposing, cognition, mechanism class, foundation models, clinical-trial prediction, allosteric modulation.

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## Introduction

Cognitive impairment across Alzheimer's disease (AD), schizophrenia (cognitive impairment associated with schizophrenia, CIAS) and Fragile X syndrome (FXS) has resisted pharmacological progress despite intensive effort. The clinical attrition landscape for cognitive enhancers is well documented: α7-nicotinic agonists (encenicline), 5-HT6 antagonists (idalopirdine, intepirdine), AMPA potentiators, and PDE9 inhibitors all reached Phase II/III and failed on cognitive endpoints, and mGluR2/3 agonists (pomaglumetad) failed across the schizophrenia programme on their primary CNS endpoints [1–5].

Computational repurposing pipelines typically prioritise candidates by one of two signals. The first is **target-binding affinity**: score how tightly a compound binds a cognition-relevant target, increasingly with large protein–ligand foundation models such as IBM's MAMMAL [6]. The second is **target genetic evidence**: weight targets by GWAS / functional-genomic support, as in Open Targets [7]. Both encode a reasonable prior — engage a relevant target — but neither was designed to answer the operative clinical question: *will a drug of this kind actually improve cognition in patients?*

Here we test the paradigms directly against real pivotal-trial outcomes, and propose an alternative: a **mechanism-class prognostic prior** that asks what the meta-analytic track record of a drug's mechanism class has been, in the relevant disease. We show this single signal dominates the target-centric paradigms, validate it three ways, repair the foundation model's documented within-target blindness, and turn the result into an actionable repurposing shortlist.

## Results

### A mechanism-class prognostic prior predicts clinical outcome; target affinity and genetics do not

We curated a leakage-audited ledger of **31 cognition drugs** with adjudicated pivotal-trial outcomes (13 approved/positive, 18 Phase II/III failures) spanning 11 mechanism classes, each with a documented effect size and citation (Methods). We defined four leakage-free predictors of clinical SUCCESS vs FAILURE and scored each against outcomes it never saw:

| Predictor | n | AUROC | 90% CI | perm p |
|---|---|---|---|---|
| **Mechanism-class track record (class leave-one-compound-out)** | 31 | **1.00** | [1.00, 1.00] | **0.0002** |
| Target genetic relevance (Bayesian Cluster-D posterior θ) | 26 | 0.59 | [0.38, 0.79] | 0.22 |
| Target binding affinity (MAMMAL DTI percentile) | 14 | 0.47 | — | 0.62 |
| Target binding affinity (original 13-target grid) | 10 | 0.12 | [0.00, 0.38] | 0.96 |

The mechanism-class predictor — for each held-out drug, the empirical-Bayes-shrunken mean effect size of its mechanism-class *siblings*, with the drug's own outcome removed — separates SUCCESS from FAILURE perfectly. As a familiar worked example, the 9 canonical Phase

III failures (encenicline, idalopirdine, intepirdine, pomaglumetad, PF-04447943, SUVN-502, ABT-126, TC-5619, MK-0249) are each predicted to fail because each sits in a failure-pure class; this is the AUROC 1.00 result restated for recognisable drugs, not a separate validation. The label-permutation test is exact: **0 of 5000 permutations achieved an AUROC ≥ the observed value** ($p = 1/5001 \approx 0.0002$). The two genuinely leakage-free target-centric paradigms remain at or below chance (Fig. 1A).

**Interpreting AUROC = 1.00 — it is a class look-up, and we quantify exactly why.** Perfect separation is not a generalisable predictive feature. It is a direct readout of **mechanism-class outcome homogeneity**, which in this ledger is complete: **all 11 mechanism classes are outcome-pure** — every class is uniformly SUCCESS or FAILURE (every AChE-inhibitor and stimulant succeeded; every 5-HT6, AMPA-PAM, PDE9 and α7 agent failed; Supplementary §1b). With pure classes, the leave-one-compound-out prior — self removed — reduces to a historical class look-up, and any predictor that respects class boundaries separates perfectly. Two consequences follow. First, the conventional drug-level bootstrap CI of [1.00, 1.00] understates uncertainty, because it cannot vary the class composition; the honest interval comes from a **class-level (cluster) bootstrap** that resamples the 11 classes themselves. We report it: it is also [1.00, 1.00] (median 1.00), not because the estimate is precise but because the classes are degenerate-by-purity — confirming that the relevant uncertainty is not sampling variance at all. Second, the binding uncertainty is therefore entirely about **out-of-sample generalisation to new mechanism classes**, which we bound directly: a leave-one-CLASS-out analysis, predicting each drug from *entirely different* classes, gives AUROC **0.00**. This 0.00 is informative — it is perfect *anti*-correlation, not randomness: removing a SUCCESS class shifts the remaining global mean toward failure (so its held-out successes are predicted to fail), and removing a FAILURE class shifts it toward success. The deeper reading is that the ledger carries **zero between-class generalisation signal** — outcomes do not transfer across mechanisms — which is the honest extrapolation ceiling. The method quantifies the historical class destiny that target-level metrics ignore; it does not forecast an unseen mechanism. That this homogeneity exists at all is plausibly a feature of cognition-drug biology — failure dominated by network-level compensation and the engagement-to-behaviour gap rather than by molecular potency.

A naïve “target popularity” baseline — the number of ChEMBL bioactivity records at the target — scored AUROC 0.96. This is a **hindsight confound**, not a predictor: a target accrues records *because* a drug succeeded there, so popularity is a downstream consequence of success. The identical confound recurs for the knowledge-graph network-propagation comparator (next section), and both are discussed as cautionary cases for graph-based repurposing (Discussion).

### The class signal is pseudo-prospective, not merely retrospective

Every drug in the ledger was curated after its outcome was known, so the strongest objection to a retrospective AUROC is that it could not have been computed *before* the trials read out. We test this directly with a temporal hold-out, exploiting the documented pivotal-readout year of each drug (Fig. 2).

**Fixed-cutoff hold-out.** Training the class prior only on the 19 drugs that read out by 2014 and predicting the 12 that read out later yields **test AUROC 1.00**. The post-2014 cognition cohort is failure-dominated (1 success, 11 failures) — the field’s real recent history — and the pre-2015 class track record ranks the single later success (pitolisant) above every subsequent failure.

**Prequential (“as-of”) evaluation.** The gold-standard design predicts *each* drug using only drugs that read out strictly before it, with no fixed cutoff. Over the 19 drugs that had at least one same-class precedent, the as-of **AUROC is 1.00** (full-coverage AUROC 0.96 over all 30 drugs with any earlier data; the 12 first-of-class drugs have no precedent and are excluded from the informed figure). Critically, the trace contains **one honest misclassification** — CX-516 (2008, AMPA-PAM), whose class's first readout had leaned slightly positive, so the as-of prediction erred *before* the AMPA class accumulated its later failures; by 2019 the same class is correctly predicted to fail. This is the method updating as evidence arrives, and it is evidence the result is not a tautology. All five 2016 α7/mGluR failures were correctly predicted from pre-2016 same-class failures — the prior *would have* flagged the cognition graveyard before it read out (Supplementary §4).

### The result is specific to the mechanism-class granularity, not an artifact of the grouping

Because the mechanism-class taxonomy is itself a curatorial choice, we test whether the perfect separation survives regrouping (Fig. 2B). Collapsing the 11 mechanism classes into 4 neurotransmitter systems — which lumps cholinesterase-inhibitor successes with α7-agonist failures under "cholinergic" — drops the AUROC to **0.62**, near chance. Randomly permuting the class labels (preserving class sizes) gives a null of **0.46 ± 0.15**, and **0 of 2000 permutations** reached the observed 1.00 (permutation $p \approx 0.0005$). The signal is therefore neither trivially robust to any grouping (coarsening destroys it, because distinct mechanisms within a transmitter system have distinct clinical fates) nor an artifact of arbitrary labels (random grouping is at chance). It is specific to the biologically-correct mechanism-class level — which is precisely the scientific claim: cognition outcomes are determined at the granularity of mechanism class.

### Out-of-sample confirmation on trials that read out after curation

The class prior also makes named, falsifiable predictions for ongoing trials, pre-registered separately with a frozen date (OSF; `data/raw/prospective_predictions.csv`). Two have read out since the ledger was frozen, both as predicted. The NMDA-coagonist-enhancement class — predicted to FAIL from the bitopertin precedent — failed twice more: **iclepertin** (GlyT1) in the Phase 3 CONNEX programme (2025; no MCCB cognition effect, programme discontinued) and **luvadaxistat** (DAAO) in Phase 2 INTERACT (2024; primary missed, development halted). Four predictions remain pending and falsifiable: PDE4 (**zatolmilast**, FXS; EXPERIENCE-301/204) and M1/M4 (**KarXT**, AD cognition; MINDSET-2) are predicted to SUCCEED, with the honest counter-signal that **emraclidine**'s M4 Phase 2 (EMPOWER) missed its psychosis primary in 2024 — recorded as a caution that tempers the muscarinic-class confidence rather than discarded. These are named drugs, trials, and endpoints with a committed prediction date, not retrospective look-ups; they are how the prior earns or loses the word *predicts* (Supplementary; `scripts/87_prospective_predictions.py`).

### The comparison is robust to predictor coverage

Each predictor is defined on a different subset of the ledger (class 31, genetics 26, affinity 14). The binding-affinity predictor requires a drug to be present in the 298-compound screening library *and* scored at its known target; the library predates the clinical ledger, so 17 mostly-obscure clinical candidates were never screened. Critically, **this missingness is structural and works against our conclusion.** Of the 17 unscored drugs, 13 are failures and 4 are successes, so the covered subset is *success-enriched* (9 SUCCESS : 5 FAILURE, 64% vs 42% overall) and consists of the best-characterised marketed drugs (donepezil, methylphenidate, memantine, pitolisant). This is the **most favourable possible test for an affinity model** — yet affinity still scores at chance. Restricting *every* predictor to this identical 14-drug common subset preserves the contrast (class **1.00**, genetics 0.53, affinity 0.47), eliminating the differing-n objection (Supplementary §2).

### The signal recovers each disease's real winning mechanism

Because mechanism-class effect sizes are disease-specific, we re-scored a differentiated (compound × target) repurposing grid using each disease's *own* pivotal-trial track record as the per-class prior, then asked which mechanism each disease elevates (Fig. 1B). The pipeline — never tuned on these diseases — recovers:

- **Alzheimer's → cholinesterase inhibitors** (the standard of care). Restricting the class predictor to the AD subpopulation ($n = 13$; 4 successes, 9 failures) gives a within-disease AUROC of **0.97** (permutation $p = 0.003$), flagging all 9 historical AD failures. As at the whole-ledger level, the 7 AD mechanism classes are themselves outcome-pure, so this 0.97 is the same homogeneity phenomenon within a single disease rather than independent evidence (the < 1.00 reflects only the singleton NMDA class, memantine, falling back to the global mean).
- **Schizophrenia (CIAS) → muscarinic M1/M4** — the mechanism of xanomeline-KarXT, FDA-approved in 2024 after decades of α7/glutamate failures.
- **Fragile X → PDE4** — the class of zatolmilast (BPN14770), positive in Phase II.

### Compared to what? Four repurposing paradigms on the identical drugs

A two-predictor comparison invites the objection that affinity and genetics are weak signals few practitioners use in isolation. We therefore evaluate four established repurposing paradigms — affinity, genetics, knowledge-graph network propagation, and chemical-structure similarity — together with an ensemble of the target-centric three, all restricted to the identical drugs on which each is defined (Supplementary §2):

| Predictor (identical 14 drugs) | paradigm | AUROC | leakage status |
|---|---|---|---|
| **Mechanism-class track record (ours)** | class history | **1.00** | leakage-audited (class-LOCO) |
| Structure nearest-neighbour to successes | chemical similarity | 0.80 | uses leave-one-out outcomes |
| Knowledge-graph personalised PageRank | network propagation | 0.80 | hindsight-confounded |
| Target genetic relevance | genetics (Open Targets) | 0.53 | leakage-free |
| Target binding affinity (MAMMAL DTI) | affinity | 0.47 | leakage-free |
| Target-centric ensemble | affinity + genetics + KG | 0.29 | mixed |

Two paradigms show apparent signal — network propagation and structure similarity, both AUROC 0.80 — but each is the kind of predictor a leakage audit exists to flag. The knowledge-graph score (personalised PageRank from the drug node to the cognition target panel over PrimeKG) rewards node degree, and a drug accrues graph edges *because* it was studied and succeeded; the structure score consumes the historical outcome labels directly. The genuinely a-priori target metrics, and even an ensemble of all three target-centric paradigms (AUROC 0.29), remain at or below chance. The most controlled contrast holds the historical information fixed: given the same record of which *other* drugs succeeded, aggregating by mechanism class (AUROC 1.00) outperforms aggregating by chemical structure (0.80).

One might worry that structure-NN is merely class membership through a chemical channel (same-class drugs share scaffolds); we checked, and the top-1 Tanimoto neighbour is in the *same* mechanism class for only **43%** of drugs — most structural nearest-neighbours are cross-class — so the structure score is a genuinely distinct, largely cross-class predictor that still underperforms class aggregation. The advantage is specific to class-level aggregation, not to having access to outcome history per se.

### The foundation model is systematically limited at allosteric/transporter interfaces; classic features, not the model, recover the ranking

Why does a 458M-parameter protein–ligand foundation model's affinity score fail to predict clinical success? In part because it cannot even rank ligands *within* a target. On a cited 21-compound allosteric benchmark, MAMMAL's predicted-pKd has a within-target standard deviation of 0.01–0.05 across ligands spanning three orders of magnitude in measured affinity; under leave-one-target-out cross-validation over **297 ChEMBL pairs (21 targets)**, its within-target Spearman ρ is **−0.12** — below chance (Fig. 1C). The model is systematically limited at the allosteric and transporter interfaces that dominate cognition targets.

A gradient-boosted learn-to-rank head over [foundation-model score, Tanimoto-to-actives, Boltz-2 3D-affinity, physicochemical descriptors], trained on ChEMBL and evaluated held-out, recovers the ranking — but a feature ablation shows the recovery is **not** the foundation model's doing. Under the 297-pair LOTO: foundation model alone $\rho = +0.06$; physicochemical alone +0.33; Tanimoto-to-actives alone +0.53; **Tanimoto + physicochemical, with the foundation model removed, +0.59**; the full fused head +0.61. Classic ligand-similarity and physicochemical descriptors do essentially all of the work; adding the 458M-parameter model and 3D-affinity lifts within-target ρ by only $\Delta\rho = +0.02$ (Supplementary). We scope this claim precisely: it concerns **within-target ligand ranking at allosteric/transporter sites using the released `dti_bindingdb_pkd` head**, a task adversarial to a sequence-only DTI model trained on BindingDB pKd, not the model's performance at its intended cross-target affinity-prediction task. For this specific repurposing sub-task, a sequence-only DTI score should not be relied upon, and inexpensive cheminformatics features suffice where it does not contribute.

### A clinician-facing repurposing shortlist

We integrate the prognostic prior, the engagement grid, the reliability flag, and GRADE-style evidence dossiers into a prospective shortlist of **approved drugs as mechanism-justified repurposing hypotheses**, restricted to success-track-record classes and flagged for novelty and safety (Fig. 1D). It surfaces literature-grounded hypotheses — for CIAS, buspirone/tandospirone (5-HT1A) and cevimeline/pilocarpine (M1/M4), with xanomeline correctly identified as already-standard; for FXS, roflumilast (approved for COPD; PDE4); for AD, fluvoxamine/blarcamesine (σ1).

**Prior-trial accounting.** Several candidates carry prior cognitive-endpoint evidence, which the shortlist treats as corroboration rather than rediscovery. Roflumilast has direct positive signals: low-dose roflumilast improved episodic memory in healthy adults and in aMCI in PDE4-cognition trials, so its appearance via the PDE4 success class is consistent with drug-level data, not merely class-level inference. Buspirone has been tested as adjunctive therapy for cognition in schizophrenia with mixed-to-null results, so its 5-HT1A hypothesis is flagged as plausible-but-unproven, not novel. Fluvoxamine's σ1 agonism has only preliminary cognition data; it is listed as exploratory. The shortlist marks each candidate's prior-evidence status; these are hypotheses to evaluate and prioritise, not predictions of efficacy, and where prior trials were null (buspirone) that is recorded against the candidate.

## Discussion

**What is, and is not, the contribution.** That a mechanism class's clinical history predicts its members' fate is, on its own, a formalisation of pharmacological common sense — and the perfect AUROC is a readout of complete class homogeneity, not a generalisable model. The contribution is the *negative* result it frames and the discipline that makes it trustworthy: under a strict leakage audit, **no a-priori target-centric predictor — affinity, genetics, network propagation, nor an ensemble of all three (AUROC 0.29) — beats chance** at forecasting cognition-drug success, and the two comparators that appear to (network propagation, structure similarity, both 0.80) do so only through hindsight that the audit exposes. The class predictor is the foil that quantifies how much signal the target-centric paradigms leave on the table; the temporal and taxonomy analyses are what license reading it as signal rather than curation. We therefore lead with the warning, not the AUROC.

The central result is uncomfortable for a field organised around molecular target engagement: in cognition, *what mechanism class a drug belongs to* — and that class's clinical history — predicts success, while how tightly it binds or how genetically-implicated its target is does not. This is consistent with a view of cognition-drug failure as dominated by biology that target-level metrics do not capture: network-level compensation, the gap between target engagement and behavioural effect, and the small effect-size ceiling reported for cognitive enhancement in healthy adults — where the best-characterised psychostimulants (modafinil, methylphenidate, d-amphetamine) reach only SMD $\approx$ 0.2–0.5 [5].

The finding reframes how foundation models should be used in this domain. A large DTI model is, at best, a coarse *engagement* gate; it is a poor *outcome* predictor and an unreliable *within-target ranker*, and our ablation shows it adds almost nothing ($\Delta\rho = +0.02$) over classic ligand-similarity and physicochemical descriptors for the latter task. Its defensible role is therefore narrow — one input gated by a class-level prognostic prior — and practitioners should not assume that a larger sequence-only model buys within-target resolution it does not have.

**A caution for graph- and target-centric learning.** The target-popularity result (AUROC 0.96) is a cautionary case for knowledge-graph and node-embedding approaches. Graph neural networks over drug–target–disease graphs implicitly reward well-connected, heavily-studied target nodes; but node degree and bioactivity-record volume are *downstream of* therapeutic success (a target is studied intensively *after* a drug works there), not predictive of it. A model that learns "popular target → viable drug" will appear strong retrospectively and fail prospectively. Our leakage audit — scoring only predictors that could have been computed before any cognition trial — is what separates the genuine signal (class track record, applied leave-one-out) from this hindsight artefact, and we recommend it as a default for repurposing benchmarks.

**Limitations.** The clinical ledger is small (n = 31), single-author-curated, and outcome-pure by mechanism class; the perfect AUROC is a homogeneity readout, and the contribution is the leakage-audited negative result on target-centric predictors (Discussion), not a generalisable model. A curated expansion to n = 42 across new mechanism classes (the AD BACE/γ-secretase/GSK3 small-molecule failures, the GlyT1 NMDA-coagonist failures, the PDE4/M1–M4 wins) leaves 17/17 classes outcome-pure (AUROC 0.995, prequential as-of 1.00; Supplementary), and two pre-registered class predictions have already been confirmed by post-curation readouts (iclepertin/GlyT1, luvadaxistat/DAAO). To address the curation-artifact concern more directly, we also drew trials from a **pre-specified, reproducible ClinicalTrials.gov query** (Phase 2/3, completed, cognition-primary, across the index indications; a 294-trial denominator across four indication×phase slices) and adjudicated every drug with a documented readout **outcome-blind** — i.e. without regard to whether it preserved purity. The resulting unbiased-sourced ledger (n = 47, 20 mechanism classes, including new GABA-B/GABA-A/filamin-A and reinforced α7/GlyT1 failure classes) remains **20/20 outcome-pure**

(AUROC 0.983; Supplementary). The binding constraint is now adjudication coverage, not author selection: the definitive remaining steps are a complete results-level adjudication of the full denominator, an independent second ledger, and orthogonal evidence axes this work does not use — such as transcriptomic connectivity-map signatures (L1000 [8]) — which we name rather than fabricate. The leave-one-class-out ceiling (AUROC 0.00) is explicit: the method cannot forecast genuinely novel mechanisms. Predicted effect sizes are bounded by the healthy-adult psychostimulant ceiling [5] applied as a conservative cap, and are enrichment rankings, not calibrated per-compound predictions. No prospective wet-lab or clinical validation has been performed; a prospective watchlist of class-based predictions for drugs in active trials is pre-registered separately (OSF). The repurposing shortlist's engagement estimates at allosteric/transporter targets are flagged uncertain per the foundation-model analysis.

## Methods

### Clinical-outcome ledger: inclusion rule, coding, and curation provenance

The ledger (`data/raw/clinical_outcomes_ledger.csv`) is a literature-curated truth set of cognition drugs with adjudicated pivotal-trial outcomes; each row carries a mechanism class, human target UniProt accession, indication, pivotal-trial identifier, readout year, the binary outcome, a pooled Hedges' $g$ on the pivotal endpoint, and a citation. No pipeline component is fit on the binary outcome.

**Inclusion rule (as applied).** A drug enters the binary analysis if it (i) was evaluated on a cognition-relevant or functional primary endpoint in its lead indication, drawn from the three index diseases (AD, CIAS, FXS) and the cognition-adjacent indications that anchor the historically successful classes (ADHD, narcolepsy/EDS); (ii) reached at least Phase II with a reported pivotal or Phase II/III readout, or obtained regulatory approval; (iii) has an assignable mechanism class and a human target UniProt; and (iv) is a small molecule within the DTI head's domain. Drugs were *excluded* if they were peptides/biologics outside the small-molecule DTI domain (e.g., GLP-1 agonists), lacked an adjudicable Phase II+ cognitive readout, or had no assignable single mechanism class/target.

**Outcome coding (pre-specified, applied uniformly).** SUCCESS = regulatory approval for, or a positive pre-registered pivotal/Phase III primary endpoint on, a cognition-relevant outcome **in the drug's lead indication**; FAILURE = a null Phase II/III primary cognitive endpoint or discontinuation for lack of cognitive efficacy. The coding is judged *within each drug's own indication* — methylphenidate is a SUCCESS on its ADHD cognitive/functional endpoint, not on AD — so the predictor learns "has this mechanism class delivered on its own pivotal cognitive endpoint," not "is this drug approved for Alzheimer's." This is the consistent rule behind every row; the per-row indication and endpoint columns make each judgement auditable.

**Assembly flow.** Of ~45 cognition/cognition-adjacent drugs reviewed, exclusions removed peptides/biologics out of small-molecule domain, agents without a Phase II+ cognitive readout, and outcomes that were ambiguous or terminated for non-efficacy reasons, leaving **31 drugs with an adjudicated binary outcome across 11 mechanism classes (13 SUCCESS / 18 FAILURE)** (a CONSORT-style flow is in the Supplementary).

**Curation caveat (honest).** This is a single-author literature curation with per-row citations, not an automated ClinicalTrials.gov extraction; a fully programmatic registry query is the obvious next step. The complete outcome-purity of the 11 classes (§ Results) could in principle be partly a curation artefact, but (a) each class's dominant clinical outcome is independently verifiable from the cited pivotal trials and reflects a pattern already well known to the field (α7, 5-HT6, AMPA-PAM, mGluR and PDE9 programmes all failed; cholinesterase inhibitors and stimulants succeeded); (b) no mixed-outcome class was split or dropped to manufacture purity; and (c) the leave-one-class-out ceiling (AUROC 0.00) is reported precisely so the result is not over-read as out-of-class generalisation.

### Predictors

**Mechanism-class prognostic prior (class leave-one-compound-out).** For each held-out drug *c*, the predicted score is the empirical-Bayes-shrunken mean clinical *g* of *c*'s mechanism-class siblings, with *c*'s own outcome removed:

$$\hat{g}\_c = (n\_sib \cdot mean\_sib + k_0 \cdot \mu\_global) / (n\_sib + k_0),$$

where mean_sib is the mean clinical *g* of the same-class drugs excluding *c*, μ_global is the ledger-wide mean *g*, n_sib is the number of siblings, and the shrinkage strength $k_0 = 1$ (one pseudo-observation pulled toward the global mean). Singleton classes (no siblings after removal) fall back to μ_global. Higher ĝ_c predicts SUCCESS. Shrinkage prevents a one-drug class from scoring at its single value and bounds the influence of small classes.

**Target genetic relevance.** σ(θ) at the drug's target, read from the V6.B "Cluster D" neurobiological posterior (`cluster_d_posterior_expanded_v2_mh8_ta99.parquet`): a hierarchical Bayesian model (numpyro NUTS) over Allen Human Brain Atlas regional expression, Open Targets Locus-to-Gene genetic scores, and single-cell specificity, with the per-target pipeline weight w_pipeline used when present and σ(θ_mean) otherwise. The posterior is fit on neurobiology and genetics only and never saw any cognition-trial outcome (leakage-free). The "MH8" variant masks substrate-mediated targets that lack a brain-expression channel; "ta99" is the reference-anchor calibration setting.

**Target binding affinity.** Within-target percentile (across the screening library) of the released MAMMAL `dti_bindingdb_pkd` head's predicted pKd at the drug's known target, from the V6.A 31-target engagement grid; leakage-free (BindingDB pKd training never saw cognition outcomes). The "original 13-target grid" row is the pre-expansion panel.

**Comparator baselines.** *Target popularity* — $\log_{10}$ total ChEMBL bioactivity records at the target. *Network propagation* — personalised-PageRank mass that diffuses from the drug node to the cognition target panel over PrimeKG [9] (`kg_ppr_sum`), summed over the panel. The target panel is the curated 31-protein cognition panel used throughout (defined by neurobiology, not by which drugs succeeded — so the panel itself does not leak outcomes); PrimeKG supplies the drug–protein–disease edges, personalised PageRank uses the standard restart probability (0.15) from the drug seed node, and `kg_ppr_sum` is the total restart-normalised mass reaching the panel. The confound is not the panel but the drug's *degree*: a drug accrues PrimeKG edges (and therefore PageRank mass) in proportion to how intensively it was studied, which is itself downstream of success. *Chemical-structure similarity* — leave-one-out maximum Morgan-fingerprint (radius 2, 2048-bit) Tanimoto from each ledger drug to any *other* ledger drug that historically succeeded; we separately report how often the top-1 structural neighbour shares the drug's mechanism class. Popularity and network propagation are reported as hindsight-confounded (not leakage-free); structure similarity, like the class prior, uses other drugs' leave-one-out outcomes and is the chemical-aggregation analogue of the class predictor.

### Metrics

AUROC via the Mann–Whitney *U* statistic with tie-averaging (numpy-only). Drug-level bootstrap CIs (2000 resamples). One-sided label-permutation p-values (5000 permutations; reported as the exact fraction of permutations meeting or exceeding the observed statistic). Paired AUROC bootstrap for head-to-head comparisons. For the class-aggregated predictor we additionally report a **class-level (cluster) bootstrap**: the 11 mechanism classes are resampled with replacement (each replicate treated as an independent cluster with its own leave-one-out siblings), the class-LOCO predictor and its AUROC recomputed per resample, and the 90% percentile interval reported — the correct uncertainty unit when the predictor is class-aggregated.

**Temporal validation.** A fixed-cutoff hold-out (train on `readout_year` ≤ T, predict later drugs, unseen-class drugs falling back to the training-window global mean) and a prequential "as-of" evaluation (predict each drug from only strictly-earlier drugs; informed predictions require ≥ 1 prior same-class sibling). **Taxonomy sensitivity** re-scores under a coarse neurotransmitter-system map and under 2000 random class-label permutations.

**Multiple comparisons.** The single confirmatory test is the class-prognostic AUROC vs the label-permutation null ($p = 0.0002$). All other quantities — the four comparator paradigms, the disease reframes, the taxonomy variants — are *descriptive contrasts* reported with their own permutation nulls or CIs, not independent confirmatory hypotheses; readers should apply the usual discount for the ~12 AUROCs tabulated (a Holm threshold at $\alpha = 0.05/12 \approx 0.004$ leaves the headline result, $p = 0.0002$, and the within-AD result, $p = 0.003$, significant; no marginal claim rests on an uncorrected p near 0.05).

### Disease reframe

Per disease $d \in \{\text{AD, CIAS, FXS}\}$, mechanism-class effect-size priors are built from the ledger plus a modulator-anchor table, restricted to the disease population and weighted by per-row evidence weight $k$ (trial-size/quality), giving a $k$-weighted mean $g$ per class within $d$. The differentiated v11 (compound × target) grid is re-scored with the disease prior and a disease-specific effect-size ceiling (AD 0.75, CIAS 0.70, FXS 0.95, reflecting the larger attainable effect in the FXS population). A within-disease class-LOCO reproduces the retrospective test inside the AD subpopulation.

### Learn-to-rank head

Gradient-boosted regressor (scikit-learn `GradientBoostingRegressor`, n_estimators = 200, max_depth = 2, learning_rate = 0.05) over features [MAMMAL pKd, Tanimoto-to-actives, Boltz-2 affinity + pose confidence, RDKit physicochemical descriptors], trained on ChEMBL pChEMBL labels. Evaluation is held-out: a cited 21-compound binding-mode benchmark and leave-one-target-out cross-validation over 297 ChEMBL ligand–target pairs (21 targets), scoring within-target Spearman $\rho$. The feature ablation (Results) re-runs the LOTO over nested feature subsets.

### Panel and compute

31 cognition targets scored with the released MAMMAL DTI head on a single consumer GPU; environment and bit-for-bit reproduction in `docs/MAMMAL_SETUP.md`.

## Figures

**Figure 1.** The four-panel synthesis (`scripts/83_flagship_figure.py`): (A) AUROC forest — only class track record predicts cognition-drug success; (B) per-disease recovery of the real winning mechanism; (C) within-target ranking at allosteric/transporter sites — classic features carry it, MAMMAL adds $\Delta\rho = +0.02$; (D) the mechanism-justified repurposing shortlist.

**Figure 2.** Pseudo-prospective and granularity-specific validation (`scripts/85_temporal_validation.py`): (A) the prequential "as-of" trace — each drug predicted from only strictly-earlier drugs (informed AUROC 1.00, one honest miss); (B) the taxonomy bracket — random 0.46, coarse 0.62, mechanism class 1.00.

## Data and code availability

All code, curated data, and figures are available under Apache-2.0 at https://github.com/pierce-lonergan/MAMMAL_Cognitive_Enhancement_Drug_Repurposing; the MAMMAL model is Apache-2.0 (IBM Research). Key reproductions: `scripts/83` (Fig. 1), `scripts/84` (robustness supplement), `scripts/85` (Fig. 2, temporal + taxonomy), `scripts/86` (calibrated constructive predictor), `scripts/87` (prospective predictions, `data/raw/prospective_predictions.csv`), `scripts/88` (ledger expansion), `scripts/89` (unbiased ClinicalTrials.gov pull + 294-trial denominator). The analysis is continuous-integration-tested and fully reproducible from the public code and data. A companion OSF pre-registration documents the analysis plan and the time-stamped prospective class-prediction commitment (two predictions already confirmed at readout).

## References

*(Records verified against canonical sources and maintained in `CITATIONS.bib`.)*


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**Supplementary material.** Ledger-assembly flow, class-purity, predictor-coverage, common-subset (six paradigms), learn-to-rank feature-ablation, temporal hold-out + prequential, and taxonomy-sensitivity analyses: `reports/manuscript_robustness.md` + `reports/temporal_validation_v1.md`. Constructive calibrated predictor (LOCO-CV Brier scores + reliability diagram): `reports/constructive_predictor_v1.md` (`scripts/86`). Pre-registered prospective predictions: `reports/prospective_predictions_v1.md` (`scripts/87`). Ledger-expansion robustness (n = 42, 17/17 classes pure): `reports/ledger_expansion_v1.md` (`scripts/88`). Unbiased ClinicalTrials.gov pull (294-trial denominator, outcome-blind adjudication, n = 47 / 20 classes pure): `reports/ctgov_pull_v1.md` + `data/raw/ctgov/` (`scripts/89`). Figures: `scripts/83` (Fig. 1), `scripts/85` (Fig. 2).